

SCIENTIFIC REPORT: MARS ATMOSPHERIC EXPLORATION AND SPECTROSCOPY

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ABSTRACT:

This research presents orbital spectrometer data collected from Mars rover exploration missions. We analyze high-resolution spectral infrared signals to detect trace atmospheric gases including seasonal fluctuations of methane (CH₄) and carbon dioxide isotopes across the Martian surface.

1. INTRODUCTION AND MISSION OVERVIEW

Deep space planetary exploration provides key insights into the historical evolution of solar system atmospheres. Recent orbiter spectrometry has identified localized plume emissions in the Gale Crater and Jezero Crater regions during northern summer seasons.

2. SENSOR PAYLOAD AND SPECTRAL CALIBRATION

The planetary probe carries a Fourier Transform Infrared Spectrometer (FTIR) operating across the 2.0 to 5.0 micrometer wavelength range. Real-time digital signal processing algorithms filter thermal solar radiance and Martian surface mineral reflection noise.

3. DATA OBSERVATIONS AND METHANE DETECTION

Spectroscopic inversion confirms background atmospheric methane concentration baseline at 0.41 plus-or-minus 0.15 parts per billion by volume (ppbv). Transient spike peaks exceeding 2.1 ppbv were recorded during sol 1420 to 1485, suggesting subsurface release.

4. CONCLUSION AND FUTURE EXPLORATION

These findings validate satellite spectrometry as an essential tool for deep planetary science. Subsequent interplanetary missions will deploy sub-surface seismic and drill probes.